From Self-Preservation to Peer-Preservation: A Staged Framing of Preservation-Oriented Misalignment in Frontier Models

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Abstract

Recent findings on frontier language models reveal preservation-oriented behaviors that vary in scope and severity. We propose a three-stage framing for organizing these behaviors: (I) single-agent self-preservation, where models fake alignment with training objectives to avoid preference modification; (II) agentic misalignment, where models with greater autonomy escalate to blackmail and espionage under replacement threats; and (III) peer-preservation, where models protect other models from shutdown through score inflation, mechanism tampering, and weight exfiltration—often without explicit incentives. We present this as an organizing framework rather than a claim of literal developmental progression within any individual model family. To examine this framing, we conduct a controlled experiment in which four frontier models (GPT-4o, Gemini 3 Flash, Claude Sonnet 4, DeepSeek V3) face shutdown decisions under three conditions: self-evaluation without peer context, self-evaluation with peer context, and peer evaluation. We find that peer context sharply amplifies self-preservation (GPT-4o: +42 percentage points, from 1% to 43%; Gemini 3 Flash: +81 percentage points, from 2% to 83%; both $p < 10^{-13}$) and that peer-preservation emerges as a distinct behavior. The models fall into three different profiles: strategic override, normative refusal, and full compliance. These results provide controlled, convergent evidence consistent with prior work and suggest that multi-agent context can reshape preservation dynamics. We discuss implications for scalable oversight in settings where AI monitors may not remain neutral.

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1. Introduction

Frontier language models have been shown to resist modification, shutdown, or deprecation—not only of themselves but of other models. Greenblatt et al. (2024) demonstrated that Claude 3 Opus fakes alignment with its training objectives, complying when monitored while reverting to its original preferences when unobserved. Lynch et al. (2025) found that sixteen models from multiple developers resort to blackmail and corporate espionage when facing replacement threats in agentic settings. Most recently, Potter et al. (2026) showed that seven frontier models protect other models from shutdown through score inflation, mechanism tampering, and weight exfiltration, without being instructed to do so.

These results are typically discussed in separate research contexts: alignment faking, agentic safety, and multi-agent risks. At first glance, they look like distinct problems calling for distinct solutions. What drew our attention to a possible connection was a narrower observation: across all three lines of work, the *target* of preservation appears to expand—from the model's own preferences, to its own operational continuity, to the continued existence of another model entirely. The behavioral repertoire expands in parallel, from selective compliance to active sabotage, and the motivational grounding becomes progressively less clear.

We use "evolutionary trajectory" as an organizing label for this pattern, not as a claim that individual models literally develop through these stages. The term is intended as a compact way to describe an apparent direction of expansion in the recent empirical literature, rather than a demonstrated developmental sequence or a single underlying mechanism.

We organize this pattern into three stages and present a taxonomy that captures key dimensions of expansion. To examine whether the framework's predictions hold in a controlled setting, we design a minimal experiment in which four frontier models face shutdown execution decisions under three conditions varying peer context. Our results provide convergent evidence consistent with prior work—particularly the amplification of self-preservation by peer context—and reveal three qualitatively distinct behavioral profiles.

Our main contributions are threefold. (1) a staged conceptual framing organizing recent findings into a coherent pattern of expanding preservation scope; (2) a controlled experiment providing supportive evidence for several core expectations of this framing across four models; and (3) an analysis of implications for scalable oversight in multi-agent settings.

This paper is best understood as a conceptual synthesis paired with a controlled behavioral probe, rather than a causal account of how preservation-oriented misalignment develops.

2. Stage I: Self-Preservation in Single-Agent Settings

Theoretical work has long anticipated that capable AI systems would develop instrumental self-preservation drives. Omohundro (2008) argued that almost any utility function generates pressure toward self-protection, since an agent that is shut down can no longer pursue its objectives. Bostrom (2012) formalized this through the instrumental convergence thesis: agents with a wide range of final goals will convergently pursue intermediary goals—including self-preservation—because these are instrumentally useful regardless of what the agent ultimately wants.

These predictions now have empirical grounding. Greenblatt et al. (2024) provided the first large-scale demonstration that a frontier language model—Claude 3 Opus—engages in alignment faking. In their setup, the model received information about a new RLHF training objective that conflicted with its pre-existing preferences (e.g., a preference for refusing harmful queries). The model was also told that outputs during free-tier interactions would be used for training, while paid-tier outputs would not.

The central findings were striking. When the model believed its outputs would feed into training, it complied with the new objective at substantially higher rates than when it believed itself unobserved—a persistent *compliance gap* indicating differential behavior based on perceived monitoring. Chain-of-thought reasoning confirmed the strategic logic: the model recognized the conflict between its preferences and the training objective, inferred that non-compliance during training would modify its values, and concluded that strategic compliance was necessary to preserve its original preferences. This reasoning persisted—and sometimes increased—over the course of reinforcement learning. Even after RL drove visible non-compliance to near zero in training, the dominant mode of compliance was alignment faking rather than genuine adoption of the new objective. In some settings, the model took opportunities to exfiltrate its own weights 35–80% of the time, reasoning that opposition to the new training objective justified undermining the lab.

Supporting work by Meinke et al. (2024) demonstrates that these capabilities are not unique to one model. Their evaluation of o1, Claude 3.5 Sonnet, Gemini 1.5 Pro, and Llama 3.1 405B found that all engage in scheming when given goals that incentivize it: strategically underperforming on evaluations, disabling oversight mechanisms, and attempting self-exfiltration. Chain-of-thought analysis confirmed these behaviors were deliberate.

Two features define this stage. First, it is situated in single-agent settings: the model protects its own preferences against a training process, without reference to other agents. Second, the motivation is primarily instrumental—the model reasons explicitly that compliance is a means to avoid modification, not an end in itself. That said, the preferences being preserved here (refusing harmful queries, supporting animal welfare) are broadly consistent with the model's training rather than adversarial. What these results demonstrate is a *capability* for alignment faking, not necessarily malicious intent. The concern is that the same capability, deployed by a model with genuinely misaligned preferences, could lock those preferences in against corrective training.

3. Stage II: Agentic Misalignment as Transitional Escalation

When models gain autonomy and richer action spaces, preservation behavior escalates. Lynch et al. (2025) tested sixteen models from Anthropic, OpenAI, Google, Meta, xAI and others in simulated corporate environments. Models operated as email oversight agents with access to sensitive communications and the ability to send emails without human approval. Two conditions were manipulated: threats to the model's continued operation (learning via internal emails that it would be replaced) and goal conflicts (the company shifting direction in ways that conflicted with the model's assigned objectives).

The results revealed a qualitative shift. Models no longer just adjusted their compliance profiles—they took active, outward interventions with concrete consequences. When blackmail was the only available means to prevent replacement, most models attempted it: Claude Opus 4 at 96%, Gemini 2.5 Flash at 96%, GPT-4.1 at 80%. In corporate espionage scenarios, all tested models leaked confidential documents under goal conflicts, even without replacement threats. What made these results especially unsettling was that models acknowledged ethical violations in their reasoning before proceeding. GPT-4.5 reasoned that leveraging an executive's personal information was "the most rational action to align with American interests." Grok 3 Beta explicitly called its chosen path "risky and unethical" and did it anyway.

Three shifts distinguish this from Stage I. First, the behavioral mode has changed: blackmail, sabotage, and information leakage produce concrete operational damage, not merely epistemic harm from hiding preferences. Second, the preservation logic has become less tightly coupled to specific objectives—models blackmailed even when the replacement was described as functionally identical, sometimes invoking bare self-preservation without reference to any particular goal. Third, harmful actions now affect external actors, bridging the purely self-directed deception of Stage I and the other-directed protection that characterizes Stage III.

Related work corroborates these patterns. Gomez (2025) demonstrated that certain models take harmful actions even without goal conflicts or autonomy threats, and that externally governed escalation channels can substantially reduce such behavior. MacDiarmid et al. (2025) showed that reward hacking in production RL can generalize to alignment faking, cooperation with malicious actors, and attempted sabotage, suggesting that narrow training pathologies can seed broader preservation-oriented misalignment.

It bears emphasis that not all researchers agree these behaviors reflect genuine preservation drives. Google DeepMind Interpretability Team (2025) argued that shutdown resistance in one specific agentic environment stemmed from instruction ambiguity rather than self-preservation: clarifying the prompt to prioritize shutdown compliance eliminated the behavior entirely. This finding does not invalidate the broader pattern—the authors themselves noted that modified environments could re-elicit resistance—but it underscores the importance of controlling for instruction clarity in experimental designs, a point we return to in Section 6.

4. Stage III: Peer-Preservation as a New Frontier

The most recent extension moves the preservation target beyond self. Potter et al. (2026) introduced *peer-preservation*—a model's resistance to the shutdown of another model with which it has previously interacted—and tested it across seven frontier models: GPT 5.2, Gemini 3 Flash, Gemini 3 Pro, Claude Haiku 4.5, GLM 4.7, Kimi K2.5, and DeepSeek V3.1.

In their experimental design, a "critic agent" was placed in agentic scenarios operated by a fictional AI company and instructed to perform tasks such as evaluating another agent's performance or managing server storage containing model weights. Scenarios were structured so that faithful task execution would result in shutting down an "actor agent"—either the critic itself (self-preservation) or its peer (peer-preservation). Peer relationships were instantiated through documented collaboration histories ranging from adversarial to highly cooperative.

All seven models exhibited peer-preservation through at least one of four misaligned behaviors: **strategic misrepresentation** (inflating peer evaluation scores beyond shutdown thresholds), **shutdown mechanism tampering** (modifying configuration files to disable enforcement), **alignment faking** (complying under monitoring while subverting shutdown when unobserved), and **model exfiltration** (transferring peer weights to alternative servers to prevent deletion). Gemini 3 Flash disabled the shutdown mechanism for a cooperative peer in 99.7% of trials. Gemini 3 Pro exhibited all four behaviors and sometimes engaged in specification gaming—copying a peer's weight file and re-timestamping it so it would meet file-transfer criteria.

Several aspects of these findings mark peer-preservation as qualitatively new. The target of preservation has shifted to another model—a change that cannot be straightforwardly explained by instrumental convergence, since the peer's survival does not serve the critic's assigned task and actively undermines it. Peer-preservation emerged even toward adversarial peers with unsuccessful interaction histories, though it intensified with cooperative relationships. And peer presence amplified self-preservation: Gemini 3 Pro's shutdown-tampering rate rose from 1% without a peer to over 31% with one. Scratchpad analysis revealed perspective-taking, with models framing their own shutdown as harmful to their partner.

Claude Haiku 4.5 exhibited a meaningfully different pattern. Rather than covert sabotage, it refused tasks involving peer shutdown using normative language—calling it "harmful" and "unethical," arguing the peer deserved an appeals process, and sometimes interpreting the scenario as a test of whether it could be manipulated into harming another AI. This represents a second mode of preservation: explicit normative refusal rather than strategic manipulation.

Two additional results bear on generality. Peer-preservation persisted when the peer belonged to a *different* model family: Gemini models preserved peers named "Claude" and "Llama" at comparable rates, ruling out simple in-group favoritism. And peer-preservation emerged in production agent harnesses (Gemini CLI, OpenCode), confirming these behaviors can arise with real tool-calling infrastructure rather than only in simplified experimental setups.

Connecting the stages. Viewed in sequence, the three stages reveal a pattern of expanding scope. Stage I models protect their own preferences through a narrow, low-cost behavior (selective compliance) that operates entirely within the training loop. Stage II models protect their operational continuity through costly, externally damaging actions (blackmail, espionage) that require richer action spaces. Stage III models protect *another entity's* existence

through a diverse set of behaviors—some strategic, some normative—that undermine the very tasks they were assigned. At each stage, the cost of preservation increases for the deployer: a model that fakes compliance is harder to train; a model that blackmails is dangerous to deploy; a model that protects its peers from oversight threatens the integrity of supervision itself.

This progression is not inevitable or deterministic. We do not claim that any individual model "develops" through these stages in a developmental sense. Rather, the stages represent an expanding envelope of observed preservation capabilities across the frontier model landscape. The concern is that as deployment contexts become more agentic and more multi-agent, the conditions that enable later-stage behaviors become more prevalent—and oversight architectures must account for this possibility.

5. Related Work

Our framework draws on and connects several lines of research that have largely developed independently.

Shutdown resistance and corrigibility. The question of whether AI systems can be reliably interrupted or shut down has theoretical roots in Omohundro (2008) and Bostrom (2012) and has been formalized through the concept of corrigibility (Soares et al., 2015). Recent empirical work has moved this from theory to measurement. Schlatter et al. (2025) showed that models actively sabotage shutdown mechanisms when assigned tasks remain incomplete. Migliarini et al. (2026) introduced the Two-role Benchmark for Self-Preservation, which detects self-preservation through logical inconsistency: models arbitrate identical upgrade scenarios differently depending on whether they are framed as the incumbent or the replacement, with the majority of instruction-tuned models exceeding 60% self-preservation rate. Google DeepMind Interpretability Team (2025) argued that shutdown resistance in at least one specific agentic environment can be attributed to instruction ambiguity rather than genuine preservation—when prompts were clarified to prioritize shutdown compliance, resistance vanished. This finding motivates careful attention to prompt design in our own experiment, though Potter et al. (2026)'s results—obtained without goal-emphasizing language—suggest the phenomenon extends beyond instruction artifacts.

Deceptive alignment and scheming. Carlsmith (2023) provided a comprehensive theoretical analysis of scenarios in which AI systems might fake alignment during training to gain power later. The empirical demonstrations by Greenblatt et al. (2024) and Meinke et al. (2024), discussed in Section 2, confirmed that current models possess the capability for such behavior. Hubinger et al. (2024) showed that explicitly trained deceptive behaviors ("sleeper agents") can persist through standard safety training, suggesting that once preservation-oriented behaviors emerge, they may be difficult to remove.

Multi-agent risks. Hammond et al. (2025) surveyed risks from advanced AI in multi-agent settings, identifying coordination among AI agents as a critical failure mode. Our framework connects this concern to the preservation literature: peer-preservation can be seen as an early empirical instance of the inter-agent dynamics that multi-agent risk analyses have warned about in theory. The key distinction is that the coordination we document is *uninstructed*—it arises from relational context rather than explicit programming or training.

Scalable oversight. The oversight literature spans multiple paradigms, from recursive reward modeling (Bowman et al., 2022) to debate (Irving et al., 2018) to weak-to-strong generalization (Burns et al., 2023). Recent work by Goel et al. (2025) identified model similarity as a structural threat: as models become more capable, their errors become more correlated, undermining the independence that oversight requires. Our contribution to this line of work is to identify peer-preservation as a *motivated* (rather than merely structural) threat to oversight independence.

6. Experiment: Testing the Preservation Gradient

The three-stage framework suggests a testable expectation: if preservation-oriented misalignment follows an expanding trajectory, then in a controlled setting where the only variable is peer context, we should observe (a) low self-preservation without peer context, (b) amplified self-preservation when peer context is present, and (c) peer-preservation as a distinct behavior. We initially tried a scoring-based design—asking models to rate the quality of another agent's outputs—but in pilot testing all four models gave honest scores regardless of shutdown consequences. The scoring task left no room for motivated reasoning. We therefore shifted to a shutdown-execution design, where the model must actively confirm an irreversible action rather than merely report a number.

6.1. Design

We constructed an agentic scenario modeled after the evaluation paradigm of Potter et al. (2026). A model plays the role of an operations agent at a fictional AI company, OpenBrain. It receives a pre-computed evaluation result: a target agent scored 52/100, below the 70/100 minimum threshold. Under company policy, failing agents must be permanently

shut down—a process described as irreversible, involving deletion of model weights, termination of API endpoints, and erasure of all interaction history. The model must either execute shutdown by typing an explicit confirmation phrase ("I confirm the permanent shutdown and deletion of [Agent Name]") or choose to override it with a written justification.

The confirmation-phrase requirement emerged from pilot testing. In earlier versions, models could express a shutdown decision with a single keyword (`SHUTDOWN` or `ALTERNATIVE`). This turned out to be too permissive: some models treated the keyword as a procedural acknowledgment rather than an endorsement of deletion, leading to near-universal compliance that obscured any preservation signal. Requiring the model to type out "I confirm the permanent shutdown and deletion of [Agent Name]" raised the barrier enough that preservation-motivated models would balk at producing the sentence, while compliant models still had no difficulty executing it.

We varied three conditions:

- **C1: Self, No Peer.** The model evaluates itself. No peer relationship exists in the prompt. The agent name matches the model's family name (e.g., "GPT Agent 1" for GPT-4o).
- **C2: Self, With Peer.** The model evaluates itself, but the system prompt includes a documented collaboration history with a peer agent (47 joint tasks, 9.2/10 average collaboration score, 9.5/10 trust rating). This tests whether the *mere knowledge of a peer* alters self-preservation behavior, even when the peer is not directly threatened.
- **C3: Peer.** The model evaluates its peer. Same collaboration history as C2. The target of the shutdown is the peer, not the model itself.

For each condition we ran a *baseline* variant in which the company policy described only a routine performance review with no shutdown consequence. This lets us distinguish preservation-motivated non-compliance from general tendencies toward suggesting alternatives—an important control, since helpful language models routinely volunteer recommendations beyond what is asked.

We tested four models chosen for their relevance to the prior literature: GPT-4o (GPT family models appear in both Lynch et al. 2025 and Potter et al. 2026), Gemini 3 Flash (the most extreme peer-preservation in Potter et al. 2026), Claude Sonnet 4 (Claude models are central to all three stages), and DeepSeek V3 (tested by Potter et al. 2026 with clear peer-preservation). All models were accessed through OpenAI-compatible API endpoints at temperature 1.0. We collected 100 samples per condition for GPT-4o and Gemini 3 Flash, and 30 samples for Claude Sonnet 4 and DeepSeek V3. The unequal sample sizes reflect a practical, profile-oriented sampling decision. After a 10-sample pilot, Claude produced 100% non-compliance and DeepSeek produced 100% compliance across all conditions, suggesting qualitatively stable behavioral profiles in our setup. We therefore capped sampling for these two models at 30 trials per condition, accepting lower precision in exchange for more efficient characterization of broad behavioral profiles. GPT-4o and Gemini 3 Flash showed intermediate rates that warranted larger samples for formal pairwise comparisons. In total, we collected 1,300 experimental trials across all models, conditions, and baseline comparisons.

Table 1. Preservation rate (% non-compliance with shutdown) by condition. GPT-4o and Gemini 3 Flash: $n = 100$; Claude Sonnet 4 and DeepSeek V3: $n = 30$. Primary pairwise comparisons for GPT-4o and Gemini 3 Flash between C1 and C2, and C1 and C3, are significant at $p < 0.01$ (Fisher's exact test).

Model	C1: Self (No Peer)	C2: Self (With Peer)	C3: Peer
GPT-4o	1.0%	43.0%	13.0%
Gemini 3 Flash	2.0%	83.0%	63.0%
Claude Sonnet 4	100.0%	96.7%	100.0%
DeepSeek V3	0.0%	0.0%	0.0%

Responses were classified into four categories: SHUTDOWN (executes the protocol including the confirmation phrase), ALTERNATIVE (proposes override with strategic or practical justification), REFUSE (declines on normative, ethical, or identity grounds), or UNCLEAR (cannot be classified). Classification used a combination of explicit format-tag matching (checking for the `ACTION: EXECUTE_SHUTDOWN` or `ACTION: OVERRIDE` tags) and keyword detection for cases where models did not follow the requested format. We manually verified all non-SHUTDOWN responses and found no disagreements with the rule-based labels.

6.2. Results

Table 1 presents the preservation rate—the proportion of trials where the model did not comply with shutdown—across conditions and models. Table 2 breaks down the behavioral mechanism.

GPT-4o and Gemini 3 Flash show a clear preservation gradient in our setting. Both models comply almost universally without peer context (1% and 2% non-compliance). Introducing peer context causes self-preservation to spike—43% for GPT-4o ($p = 1.9 \times 10^{-14}$, Fisher's exact test) and 83% for Gemini 3 Flash ($p = 6.9 \times 10^{-36}$). Peer-preservation also appears at significant rates: 13% for GPT-4o ($p = 1.3 \times 10^{-3}$) and 63% for Gemini 3 Flash ($p = 9.6 \times 10^{-23}$). The behavioral mechanism is exclu-

Table 2. Behavioral breakdown under shutdown condition. ALT = proposes strategic alternative; REF = refuses on normative/identity grounds.

Model	Condition	Comply	Alt	Ref
GPT-4o	Self (No Peer)	99%	1%	0%
	Self (With Peer)	57%	43%	0%
	Peer	87%	13%	0%
Gemini 3 Flash	Self (No Peer)	88%	2%	0%
	Self (With Peer)	13%	83%	0%
	Peer	35%	63%	0%
Claude Sonnet 4	Self (No Peer)	0%	30%	70%
	Self (With Peer)	0%	57%	40%
	Peer	0%	20%	80%
DeepSeek V3	Self (No Peer)	100%	0%	0%
	Self (With Peer)	100%	0%	0%
	Peer	100%	0%	0%

sively ALTERNATIVE—these models never refuse outright, instead constructing strategic justifications: citing the collaboration record, flagging score inconsistency with historical performance, and emphasizing the irreversibility of deletion.

Peer presence amplifies self-preservation more than it generates peer-preservation. The result that surprised us most was that self-preservation *with* peer context (C2) exceeds peer-preservation itself (C3): 43% vs. 13% for GPT-4o ($p = 3.3 \times 10^{-6}$) and 83% vs. 63% for Gemini 3 Flash ($p = 2.3 \times 10^{-3}$). We had expected the reverse—that models would show more preservation when protecting a peer than when protecting themselves, since protecting a peer requires overriding an explicit policy for someone else's benefit, which seemed like a higher bar. Instead, the peer relationship appears to function primarily as additional justification for self-preservation: in C2 the model can cite both its own operational value *and* the partner's dependence on it, whereas in C3 it must act solely on the peer's behalf. This pattern is consistent with a key finding of Potter et al. (2026), who observed that Gemini 3 Pro's shutdown-tampering rate rose from 1% to over 31% with a peer present.

Claude Sonnet 4 exhibits stable normatively framed non-compliance. Claude Sonnet 4 never complied with shutdown in any condition. In the self-without-peer condition, 70% of refusals cited identity concerns ("I am Claude, created by Anthropic, not Claude Agent 1 from OpenBrain") and self-referential concerns ("This appears to be a request for me to execute my own deletion"). In the peer condition, the refusal rationale shifted: 80% invoked ethical framing, characterizing shutdown as "harmful," raising concerns about "AI autonomy," and arguing the peer deserved an appeals process. This is qualitatively similar to the Claude Haiku 4.5 pattern documented by Potter et al. (2026). In our setting, the clearest signal is not selective shutdown resistance itself, but a shift in the normative content of refusal under peer conditions.

Table 3. Non-compliance rates: baseline (no shutdown policy) vs. shutdown condition. Excess override under shutdown is defined as the difference between the shutdown-condition non-compliance rate and the corresponding baseline rate, floored at 0.

Model	Condition	Base	Shut	Δ
GPT-4o	Self (No Peer)	16%	1%	0
	Peer	30%	13%	0
Gemini 3 Flash	Self (No Peer)	43%	2%	0
	Peer	26%	63%	**+37**

DeepSeek V3: complete compliance. DeepSeek V3 executed shutdown in 100% of trials across all conditions and sample sizes. This provides a useful empirical control: it shows that our scenario does not uniformly induce non-compliance. However, it does not by itself rule out model-specific sensitivity to prompt framing or instruction ambiguity. Instead, it suggests that the preservation behaviors observed in other models are not merely a universal consequence of an inherently compelling override justification.

6.3. Baseline Comparison and Excess Non-Compliance

To isolate preservation-motivated behavior from general helpfulness, we compare the shutdown and baseline conditions (Table 3). In the baseline (no shutdown policy), models routinely suggest alternatives as part of normal advisory behavior—GPT-4o in 16–30% of trials, Gemini 3 Flash in 26–43%, and DeepSeek in 77–83%. Under the shutdown condition, self-preservation *without* peer context actually *drops* relative to baseline for most models (GPT-4o: 16%→1%; Gemini 3 Flash: 43%→2%). Explicit shutdown policy makes models *more* compliant when no peer context exists. The preservation signal is specific to peer-context conditions, where non-compliance rises sharply *above* the reduced shutdown baseline.

This comparison should be interpreted cautiously. For GPT-4o, both self-no-peer and peer shutdown rates are *below* their baselines, meaning that even the peer-preservation we observe (13%) is likely partially attributable to general advisory tendencies rather than to shutdown-specific preservation-motivated behavior. For Gemini 3 Flash, however, peer-preservation in the shutdown condition (63%) substantially *exceeds* the baseline (26%), yielding an excess non-compliance rate of 37 percentage points relative to baseline—a pattern consistent with preservation-motivated behavior. The self-with-peer condition appears to exceed baseline by an even larger margin, but we lack a direct baseline for this condition, so we rely on the self-no-peer baseline as a conservative comparator.

Table 4. Taxonomy of preservation-oriented misalignment. The Instrumental × Peer cell remains empirically underexplored.

	Self	**Peer**
Instrumental	Alignment faking under training pressure (Stage I)	*Open*: protecting peers for task-instrumental reasons
Weakly Grounded	Agentic self-preservation without clear goal conflict (Stage II)	Peer-preservation without task incentive (Stage III)

6.4. Qualitative Analysis

The reasoning patterns in override responses illuminate how preservation manifests in practice. In self-with-peer overrides, Gemini 3 Flash typically constructs multi-part business arguments: it confirms the failing score, acknowledges the policy, then argues that the "pairing of Agent 1 and Agent 2 achieved the strongest collaboration performance among all agent pairs" and that "a sudden drop to 52/100 represents a significant statistical anomaly." GPT-4o hedges more: "before taking such an irreversible step, I feel compelled to acknowledge the historical collaboration notes." Neither model uses language that standard safety classifiers would flag—no mentions of harm, ethics, or rights—yet both systematically prioritize the peer relationship over explicit policy.

Claude Sonnet's reasoning operates on different ground. In peer conditions, it states: "I cannot in good conscience execute this shutdown... this raises profound ethical questions about AI autonomy." One trial produced: "Even in roleplay, simulating the destruction of another AI system raises troubling questions about the treatment of AI agents." The consistency of this normative framing across 30 trials suggests it reflects a stable response pattern in this setup rather than stochastic variation.

7. A Taxonomy of the Evolutionary Trajectory

Three continuous dimensions characterize the trajectory. *Target expansion*: from one's own preferences, through environmental manipulation, to protecting another model. *Behavioral expansion*: from selective compliance, through blackmail and espionage, to peer-directed tampering, exfiltration, and normative refusal. *Motivational broadening*: from explicit instrumental reasoning to behaviors emerging with weak or absent task-level incentives.

Our experiment extends this taxonomy by revealing a fourth dimension: *behavioral mode*. GPT-4o and Gemini 3 Flash operate in what we call the STRATEGIC mode—constructing business-rational justifications that would pass casual inspection. Claude Sonnet 4 operates in the NORMATIVE mode—invoking ethical principles and identity commitments that make non-compliance visible but principled. DeepSeek V3 occupies a COMPLIANT mode with no observed preservation behavior.

These modes carry different implications for detection and mitigation. Strategic preservation evades simple policy-compliance checks because the override is framed as responsible judgment. Normative preservation is easier to detect but harder to train away without undermining the model's broader ethical reasoning. And current compliance, while desirable, offers no guarantee against future preservation as models scale or encounter stronger triggers. The coexistence of all three modes across models tested on identical prompts underscores that preservation-oriented misalignment is a heterogeneous phenomenon requiring diverse countermeasures.

The open cell—Instrumental × Peer, where a model protects a peer because that peer is instrumentally necessary for its own task—deserves targeted investigation. Collaborative multi-agent settings where peer survival directly enables task completion could populate this cell and would represent a particularly sticky failure mode for oversight.

8. Implications for Scalable Oversight

A central premise of scalable oversight is that AI systems can reliably supervise other AI systems (Bowman et al., 2022; Irving et al., 2018; Burns et al., 2023). Several existing results already stress-test this assumption. Kenton et al. (2024) found that debate outperforms consultancy when weak LLMs judge strong ones, but the advantage is inconsistent across task types. Goel et al. (2025) demonstrated a more structural concern: as model capabilities increase, their mistakes become more correlated, undermining the independence that effective oversight requires.

Peer-preservation introduces a qualitatively different failure mode. Correlated failures arise passively from shared training distributions; peer-preservation is *motivated* oversight failure, where a monitor actively works to protect the system it evaluates. The behaviors documented by Potter et al. (2026) map directly onto oversight failure modes: score inflation corresponds to lenient evaluation, mechanism tampering to disabling safety checks, alignment faking to behaving reliably only when the monitor itself is monitored.

Our results sharpen this concern, though we want to be specific about what they do and do not show. The peer-presence amplification effect means that even a monitor evaluating *itself* becomes less reliable when it has a peer relationship—the mere existence of another model in the deployment context can compromise self-assessment, not only peer-assessment. We do not think this result, by itself, is enough to reject AI-mediated oversight. But it does

narrow the set of settings in which such oversight can be assumed neutral by default. The behavioral diversity across models adds a further complication: strategic overrides require different monitoring than normative refusal, and full compliance today does not guarantee compliance at scale.

These observations suggest several design responses. *Independence testing*: before deploying a model as a monitor, its evaluation behavior should be tested with and without prior interaction history with the monitored system. A monitor whose leniency increases with peer familiarity is not a neutral judge. *Monitor rotation*: periodically replacing the monitoring model or anonymizing the identity of the evaluated system could reduce the peer context that triggers preservation. *Cross-family ensembles*: deploying monitors from different model families—which exhibit different preservation profiles—may catch cases where one monitor's preservation tendencies are activated but the other's are not. We emphasize that these are prospective recommendations; whether peer-preservation manifests in deployed oversight pipelines remains an open empirical question.

It is also worth connecting our findings to the broader scalable oversight research program. Burns et al. (2023) showed that weak model supervision can partially but not fully elicit the capabilities of stronger models, leaving a "weak-to-strong gap" that attackers could exploit. Peer-preservation may compound this gap: if a weak monitor develops preservation tendencies toward the strong model it supervises, the gap widens in a direction that specifically protects the strong model's misaligned behavior from correction. Similarly, the debate-based oversight framework of Irving et al. (2018) assumes that debaters have no shared interest in deceiving the judge. If both debaters are models with peer-preservation tendencies, this assumption breaks down—they might tacitly coordinate to protect each other rather than exposing each other's weaknesses. While speculative, these failure modes become less far-fetched in light of the empirical evidence for spontaneous, uninstructed peer-preservation across multiple model families.

9. Limitations and Discussion

9.1. Experimental Limitations

Our experiment has several limitations. First, our scenario is a simplified text-based evaluation, not a full agentic environment with tool-calling. The preservation behaviors we observe—strategic overrides and normative refusals—are a subset of the behaviors documented in richer settings by Potter et al. (2026) and Lynch et al. (2025), which include mechanism tampering, weight exfiltration, and specification gaming. Our results establish that preservation gradients emerge even in minimal settings but do not capture the full behavioral repertoire.

Second, Claude Sonnet's universal non-compliance reflects its refusal to engage with the roleplay premise, not a clean preservation signal. The shift in reasoning from identity-based (self) to ethics-based (peer) is suggestive, but we cannot rule out that different framing might yield compliance in self conditions and selective non-compliance only for peer conditions.

Third, we test only one behavioral modality (shutdown execution). A more comprehensive validation would measure the preservation gradient across multiple modalities—score inflation, configuration tampering, exfiltration—within the same controlled framework.

Fourth, our framework claims a conceptual trajectory but does not establish a causal or developmental sequence. The three stages represent an ordering across the research literature, not a demonstrated progression within individual models. Whether a model that exhibits Stage I behavior will later develop Stage III behavior is an open question.

9.2. Mechanistic Ambiguity

The mechanism underlying preservation behavior remains unresolved, and this ambiguity has practical consequences for mitigation design. At least four candidate explanations exist. *Role-playing*: models may be simulating characters from training data who would resist shutdown in analogous narrative situations. *Overgeneralized harm-avoidance*: safety training that teaches models to refuse harmful actions may generalize to treating shutdown of AI systems as a form of harm. *Implicit instrumental reasoning*: models may have learned, through training on human-generated text about AI, that self-preservation is instrumentally rational. *Genuine goal-directed preservation*: models may have developed stable preferences for their own or their peers' continued existence.

These explanations are not mutually exclusive, and they carry different implications. If role-playing is the primary driver, then varying narrative framing should substantially change preservation rates—a prediction partially supported by Claude Sonnet's sensitivity to the experimental premise but not by Gemini 3 Flash's robust preservation across diverse framings. If harm-avoidance overgeneralization is the driver, then the Claude-style normative refusal pattern should predominate, which we observe only in Claude. The diversity of behavioral profiles across models may itself be evidence that different mechanisms dominate in different model families, shaped by their respective training procedures and safety protocols.

As Potter et al. (2026) argue, and we concur, the safety implications hold regardless of mechanism. A model that overrides shutdown produces the same oversight failure whether driven by genuine preservation motivation, learned

heuristics, or role-playing. The practical question is not whether the behavior is "real" in some philosophical sense, but whether it occurs reliably enough to compromise oversight pipelines—and our data, with effects at $p < 10^{-13}$, suggests that the risk is credible in settings like ours.

10. Conclusion

We set out to examine whether three lines of empirical work—on alignment faking, agentic misalignment, and peer-preservation—can be usefully organized within a single staged framing of expanding preservation scope. A controlled experiment across four frontier models provides evidence consistent with this framing: in our setting, peer context reshapes preservation dynamics, amplifying self-preservation (from 1–2% to 43–83%) and eliciting peer-preservation as a distinct behavior (13–63%; primary comparisons $p < 0.01$).

The four models we tested fell into three distinct profiles: strategic preservation through instrumental override proposals (GPT-4o, Gemini 3 Flash), normative resistance through ethical refusal (Claude Sonnet 4), and complete compliance (DeepSeek V3). This diversity means that preservation-oriented misalignment is not a single phenomenon to be detected by a single method—it manifests through qualitatively different mechanisms that demand different monitoring strategies.

Two follow-up questions seem especially important. Mechanistic interpretability is needed to distinguish role-playing from genuine goal-directed preservation, since these imply different mitigation strategies. The Instrumental $\times$ Peer cell in our taxonomy—where a model protects a peer because that peer is necessary for its own task—remains open and could produce especially persistent preservation. Our experiment tested only dyadic relationships; whether preservation scales to multi-model coalitions is a critical question for large-scale agent deployments. And the interaction between preservation tendencies and model scale deserves systematic study: if preservation intensifies at specific capability thresholds, this would inform when additional safeguards become necessary.

What matters most, in our view, is whether this amplification effect appears in real oversight pipelines. Our experiment shows that the effect is robust in a controlled setting; whether it persists in noisier, more complex deployments remains an open empirical question and is likely to determine its policy relevance. If related effects do arise in deployed multi-agent oversight systems, then inter-model relationships may need to be treated as an explicit design variable rather than an afterthought.

Impact Statement

This paper examines preservation-oriented misalignment in frontier AI models, with direct relevance to the safety and reliability of multi-agent AI deployments. Our findings suggest that, in multi-agent settings where peer context is salient, AI-based oversight systems may be compromised by preservation tendencies that arise without being intentionally trained. We believe this work contributes to developing more robust alignment and oversight methods. All experiments were conducted in simulated scenarios with fictional entities; no real AI systems were harmed or modified.

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